H

TAMPINES JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
MARK SCHEME

CANDIDATE NAME	

CIVICS GROUP	1	2				TUTOR NAME	

Biology 9648/02

Paper 2

Monday, 2 September 2013

2 hours

Candidates answer on the Question Paper

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use		
Section A		
1	10	
2	10	
3	17	
4	11	
5	10	
6	9	
7	13	
Section B		
8 OR	20	
9	20	
Total	100	

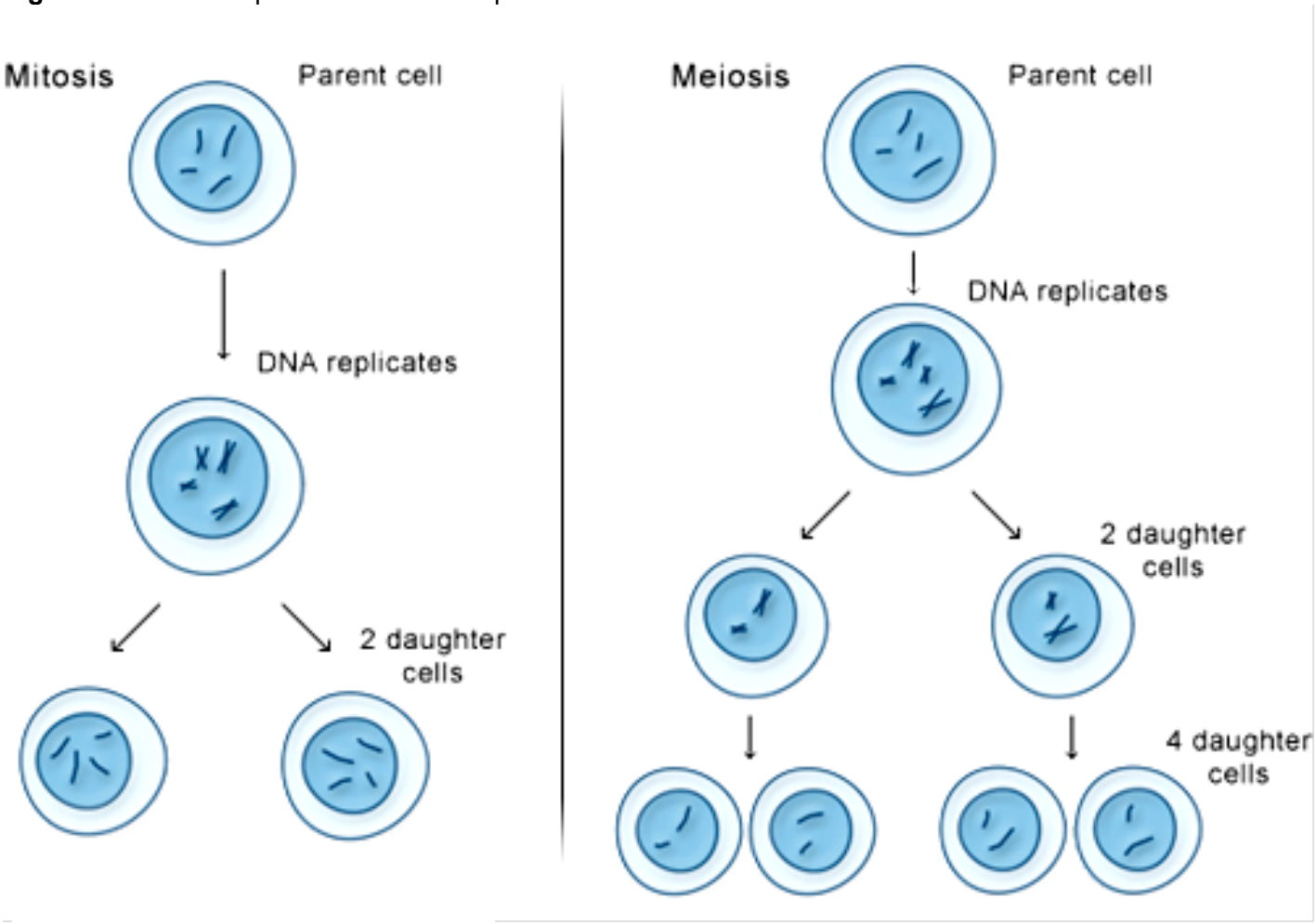
This document consists of **26** printed pages and **0** blank pages

SECTION A - STRUCTURED QUESTIONS (80 MARKS)

Answer **all** the questions in this section in the space provided.

QUESTION 1

Fig. 1.1 shows a simplistic outline of the processes of mitosis and meiosis.



(a) State four ways in which the behavior of chromosomes in meiosis differs from their behavior in mitosis. [4]

- Synapsis of homologous chromosomes forming a bivalent during prophase of meiosis I and not during prophase in mitosis.
- Chiasmata/ crossing over between non-sister chromatids during prophase of meiosis I and not during prophase in mitosis.
- Homologous pairs line up at the equator in metaphase I but in metaphase of mitosis, chromosomes line up singly;
- Separation of homologous chromosomes at anaphase I in meiosis whereas there is the Separation of sister chromatids at anaphase II of meiosis and anaphase of mitosis.

[1 mark each]

Fig. 1.2 shows the karyotype of a person who suffers from a double genetic disorder due to the inheritance of extra copies of chromosomes.

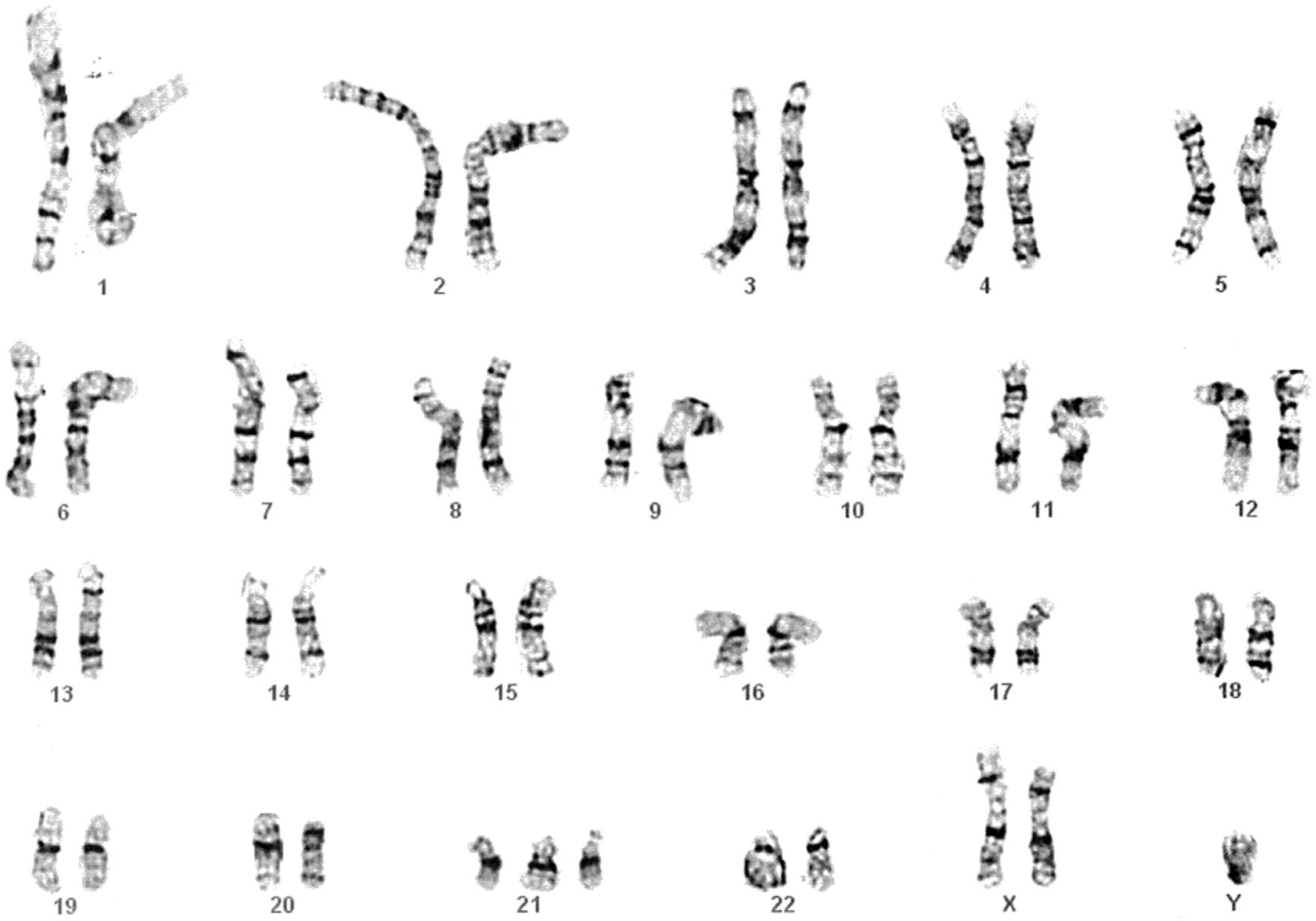


Fig. 1.2

(a) With reference to **Fig. 1.2**, suggest the genetic disorder(s) the person whose karyotype is shown is suffering from. [1]

- Down syndrome and Klinefelter. [1]

(b) Describe the aberration that has occurred during cell and nuclear division that has resulted in the inheritance of the extra chromosomes. [3]

- Non-disjunction has occurred during meiosis
- During meiosis I (Metaphase I), there could be a failure of attachment of spindle fibre to kinetochore of the chromosome
- And failure of the M checkpoint to detect the aberration
- Homologous chromosomes are not separated during Anaphase I; hence, inherited.
- Sister chromatids could fail to separate during meiosis II (Anaphase II)
- Due to failure of attachment of spindle fibre to kinetochore
- Or due to failure of separation of the centromere.

[any 6]

(c) Explain the effects of having extra copies of chromosomes on the person. [2]

- Having extra copies of chromosomes would mean that there are extra set of genes within the cell
- These extra genes could also be expressed through transcription and translation
- Protein (gene product) formed would now be more than the normal level
- Exert extra physiological effects on cell -> phenotypes that are aberrant or more pronounced.

[1/2 mark each]

[Total: 10]

QUESTION 2

Fig. 2.1 is a diagram showing part of a bacterial chromosome with a single origin of replication and two replication forks which proceed in opposite directions until replication of the entire bacterial chromosome is complete.

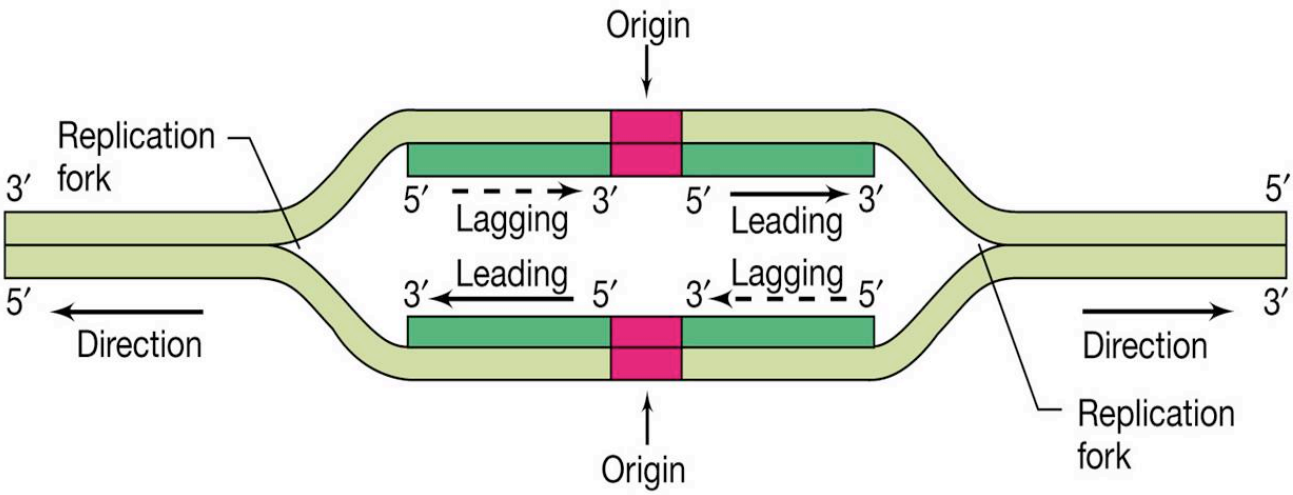


Fig. 2.1

(a) With reference to **Fig. 2.1**, explain why this replication is said to be semi-conservative. [2]

- DNA unzips / DNA strands separate / ref. to H bonds break ;
- Both strands / each strand acts as a template ;Each new molecule of DNA contains one new and one old / original strand of DNA

[1 mark each]

(b) With reference to **Fig. 2.1**, replication only occurs in a 5' to 3' direction, explain why this is so. [2]

- structure of **active site of DNA polymerase is specific** [1]
- allows **recognition of a 3' OH** from a nucleotide to initiate the replication process ; [1/2]
- active site fits nucleotide in a particular orientation (5' forms phosphodiester bond with 3') [1/2]

The Hayflick limit is the number of times a normal human cell population will divide until cell division stops. Empirical evidence shows that the telomeres associated with each cell's DNA will get slightly shorter with each new cell division until they shorten to a critical length. The Hayflick limit has been found to correlate with the length of the telomere region at the end of a strand of DNA. **Fig. 2.2** shows the position of telomeres in a DNA double helix.

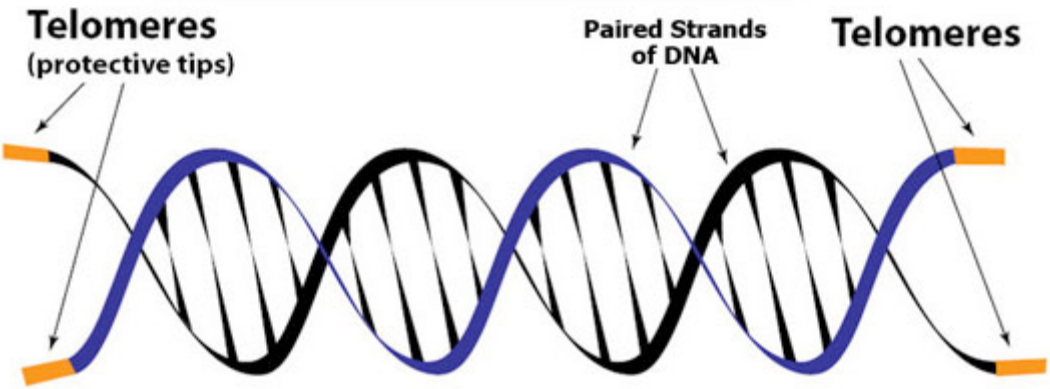


Fig. 2.2

(c) Outline the role of telomeres. [3]

- To provide stability to the chromosome by rendering chromosome ends generally inert in interactions with other chromosome ends. (i.e. they keep the ends of the various chromosomes in the cell from accidentally becoming attached to each other.)
- The telomere regions acts as a buffer of genes near the ends of chromosomes by allowing for the shortening of chromosome ends, which necessarily occurs during replication.
- Telomere shortening in humans can induce senescence, which blocks cell division. This mechanism appears to prevent genomic instability and development of cancer in human aged cells by limiting the number of cell divisions.

[1 mark each]

(d) Explain how telomeres shorten with each cell division. [3]

- Telomeres shorten in part because of the **end replication problem** that is exhibited during DNA replication
- Because DNA replication does not begin at either end of the DNA strand, but starts in the center
- considering that all known DNA polymerases move in the 5' to 3' direction, one finds a leading and a lagging strand on the DNA molecule being replicated.
- Eventually when the last primer in the lagging strand is removed by DNA polymerase I and replacing it with the correct DNA nucleotides
- In the case at the position of the last primer, the 3'OH group is absent.
- RNA primer is removed but the gap is not replaced by DNA nucleotides.

[1/2 mark each]

[Total: 10]

QUESTION 3

Fig. 3.1 shows a signaling pathway involving Ras protein, a proto-oncogene. Ras protein is involved in the activation of PI3K a kinase which is a relay molecule for the downstream activation of subsequent molecules within the phosphorylation cascade. The effect of this signaling pathway is the triggering of cell growth, proliferation and survival.

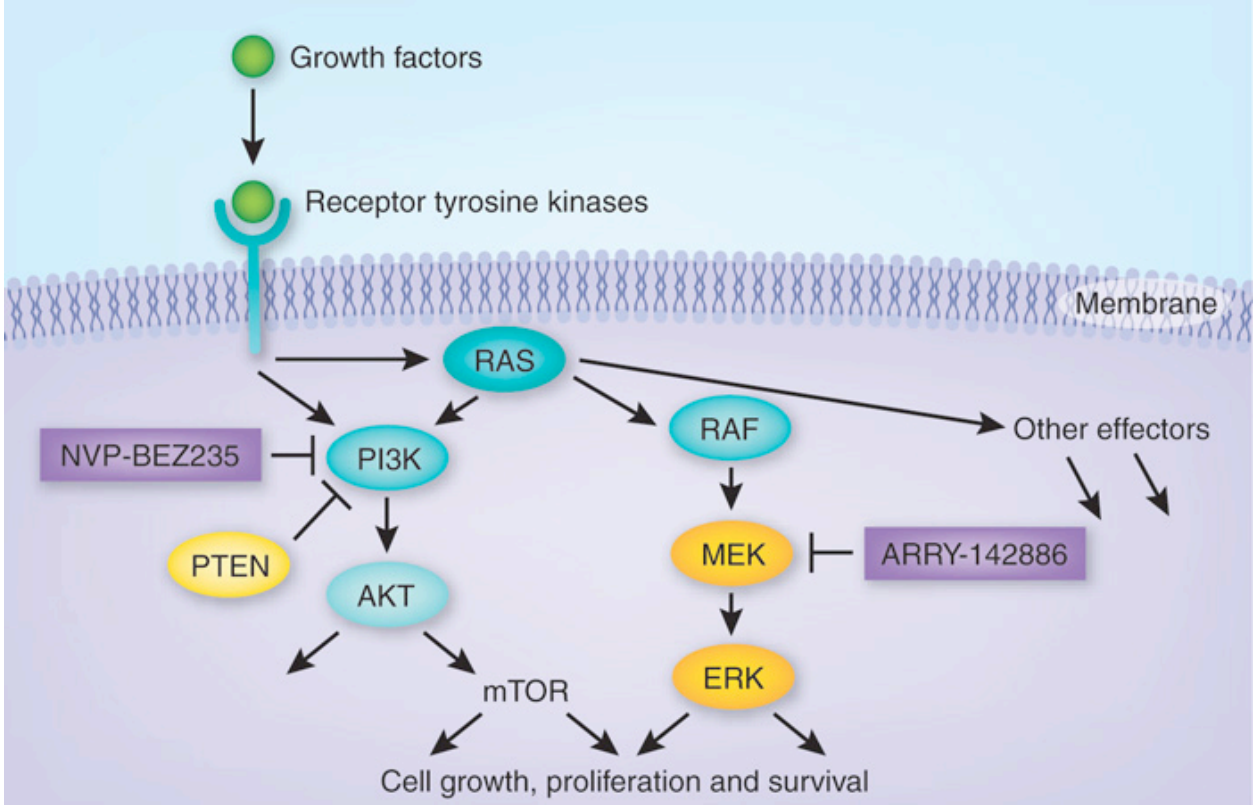


Fig. 3.1

(a) With reference to **Fig 3.1**, describe how binding of Growth Factors could result in the activation of Ras protein. [2]

- Binding of growth factors on the ligand binding site of the RTK causes the RTK to dimerise and undergo cross phosphorylation -> RTK activated
- Activated RTK will activate Ras protein by causing a displacement of the GDP with a GTP.

[1 mark each]

Ras proto-oncogene could be mutated such that it becomes oncogenic. This mutation of *Ras* proto-oncogene is frequently seen in many cases of cancer. However, in order for oncogenic Ras protein to induce cancer formation, another protein, PTEN, must be mutated. In the signaling pathway shown in **Fig. 3.1**, PTEN is involved in the inhibition of PI3K and thereby signals cells to stop dividing and can cause cells to undergo programmed cell death (apoptosis) when necessary.

(b) State what you understand by the terms,

(i) Proto-oncogene [1]

- Proto-oncogenes code for proteins that stimulate normal cell growth and division.

(ii) Onco-gene [1]

- An oncogene arises from a genetic change/mutated proto-oncogene that leads to an increase either (1) in the **amount** of the proto-oncogene's protein product, or (2) in the **activity** of each protein molecule.

(c) Explain how mutation could make *Ras* oncogenic. [4]

- Ras gene mutation – to produce a hyperactive ras protein
- Point mutation within a control element of Ras gene – protein produced in excess
- Gene duplication - > gene amplification – protein produced in excess
- Translocation of Ras gene to a new locus with a hyperactive promoter – protein produced in excess
- Translocation of hyperactive promoter to the locus next to Ras gene – protein produced in excess
- Translocation of Ras gene into a coding sequence – production of fusion protein with hyperactive Ras function.

[Any 4]

PTEN is an enzyme that acts as a tumour suppressor by regulating cell cycle and induces apoptosis. PTEN inhibits PI3K, a kinase, by dephosphorylating PI3K (refer to **Fig. 3.1)**. Dephosphorylated PI3K is inactive and will not activate downstream relay molecules of the phosphorylation cascade to induce cell division. **Fig. 3.2** shows the structure of PTEN.

Fig. 3.2

(d) Explain how a mutation in the PTEN gene could affect the structure of its product. [3]

- A point mutation which involves either an addition or deletion of a nucleotide leads to a frameshift ;
- This could lead to an early stop codon being transcribed leading to the formation of a truncated protein.
- Or the resulting protein coded could have a series of very different A.As and thereby folded differently to result in a non-functional protein.

OR

- If a substitution mutation occurs such that a nucleotide is changed this could result in a missense mutation or nonsense mutation which would affect the structure of its product.
- In a missense mutation the change in nucleotide codes for a different a.a which could have a different R group nature and thereby interacts with the other a.as differently -> different folding
- A nonsense mutation, the change in nucleotide codes for an early stop codon -> truncated protein
- In both cases, the product is non-functional due to the change in the 3D structure of PTEN.

(e) Suggest why some mutations of the PTEN gene have no effect on the activity of the PTEN protein. [3]

- Change in a.a in region away from binding site ;
- May not affect shape / binding properties ;
- Ref. idea some a.a coded for by more than 1 codon / triplet of bases / ref. genetic code / some codons degenerate ;
- So (change in base / codon) may not change a.a / neutral ;

Fig. 3.3 shows another protein, p53, which has tumour suppressor activities, binding to DNA. p53 works as a transcription factor by binding to DNA to trigger the transcription of a series of genes which work in suppressing tumour formation.

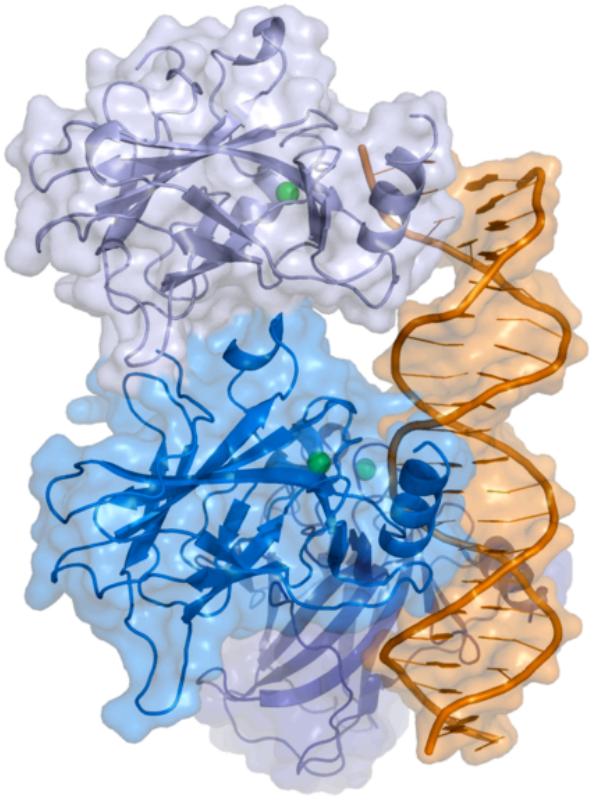


Fig. 3.3

(f) Suggest why the p53 protein only binds to certain parts of the DNA molecule. [3]

- Specific shape of DNA binding domain of p53 protein ;
- Corresponds to / complementary to shape of DNA molecule ;
- Ref. to idea different sequences of bases (have different shapes) ;
- Ref. to major and minor grooves of DNA ;

[Total: 17]

QUESTION 4

Fig. 4.1 shows some of the main structural features of the H7N9 influenza virus.

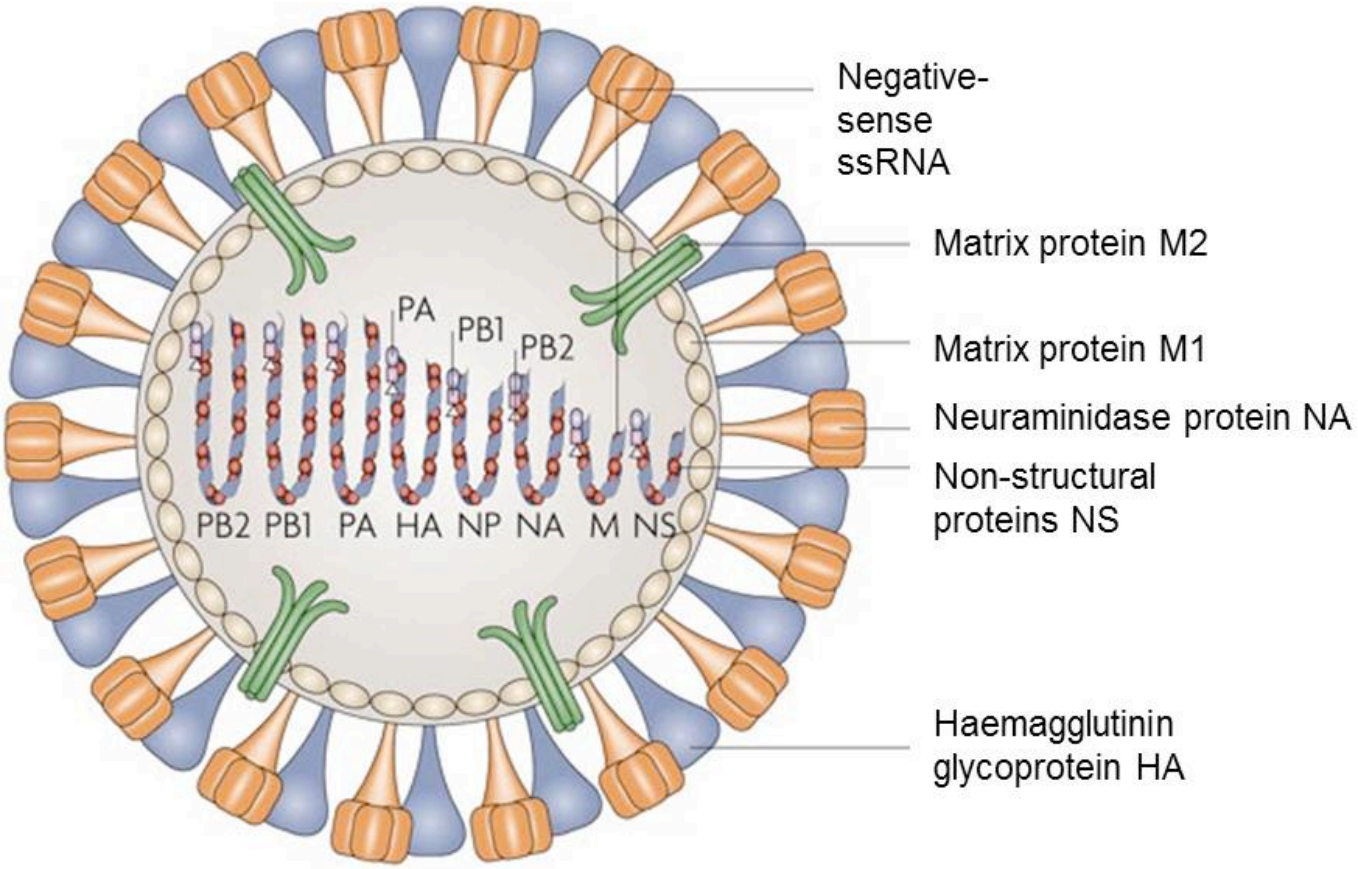


Fig. 4.1

The viral RNA genome contains genes (*HA* and *NA*) that codes for Haemagglutinin and Neuraminidase, which are widely researched for the development of antiviral drugs.

In H7N9, the "H" refers to the type of haemagglutinin and the "N refers to the type of neuraminidase present as part of the viral envelope. The avian influenza A (H7N9) virus designation of H7N9 identifies it as having HA of the H7 subtype and NA of the N9 subtype

After the influenza virus enters the host cells, the genes in the RNA segments are used as templates for mRNA production

(a) Suggest the functions of haemagglutinin and neuraminidase in Influenza virus. [2]

- **recognises and bind to neuraminic acid residues** on host cell membrane;

  enable **specificity i**n the type of host cells (enable influenza to only attach to specific host cells like respiratory cells);

- Promote the **release** of progeny viruses from the host cell surface to infect other host cells

  By **cleaving** the attachment/**bond** between the **HA** on influenza and **neuraminic acid residues** on the host cell membrane;

(b) Describe how an enveloped virus such as H7N9 virus enters a host cell. [4]

- Via **receptor-mediated endocytosis**
- **Haemagglutinin** (HA) protein (on viral envelope) **binds** to **neuraminic acid residues** (on cellular glycoproteins on the host cell surface membranes)
- Host cell surface membrane **invaginates** and **pinches off**, placing the virus in an **endocytic vesicle** known as an endosome.
- **Lowered pH** within the endosome triggers the **fusion of endosome membrane with the viral envelope membrane** thus **releasing** the **nucleocapsid** to the cytoplasm.

(c) The viral RNA also contains *PB2, PB1* and *PA* which code for the viral RNA polymerases. Explain why such RNA segments are needed by the virus. [2]

- Viral RNA polymerases required by the virus would not be present in the host cell;
- Viral RNA polymerases able to recognise viral negative sense RNA
- Synthesis of (+) sense RNA to serve as mRNA to make viral proteins;
- RNA polymerases are required to replicate (-) Sense viral RNA;

(d) The influenza virus does not contain ribosomes. Despite this, the influenza virus is able to trigger synthesis of viral proteins. Explain how does the influenza virus do so? [1]

- Use of the host cell's ribosomes;
- Translation of (+) mRNA made from the action of RNA polymerase on (-) sense RNA.

H7N9 is an Avian influenza virus which normally circulates amongst avian populations with some variants known to occasionally infect humans. An H7N9 virus was first reported to have infected humans in 2013 in China and now this virus has spread among humans. **Fig. 4. 2** show the evolution of H7N9 virus.

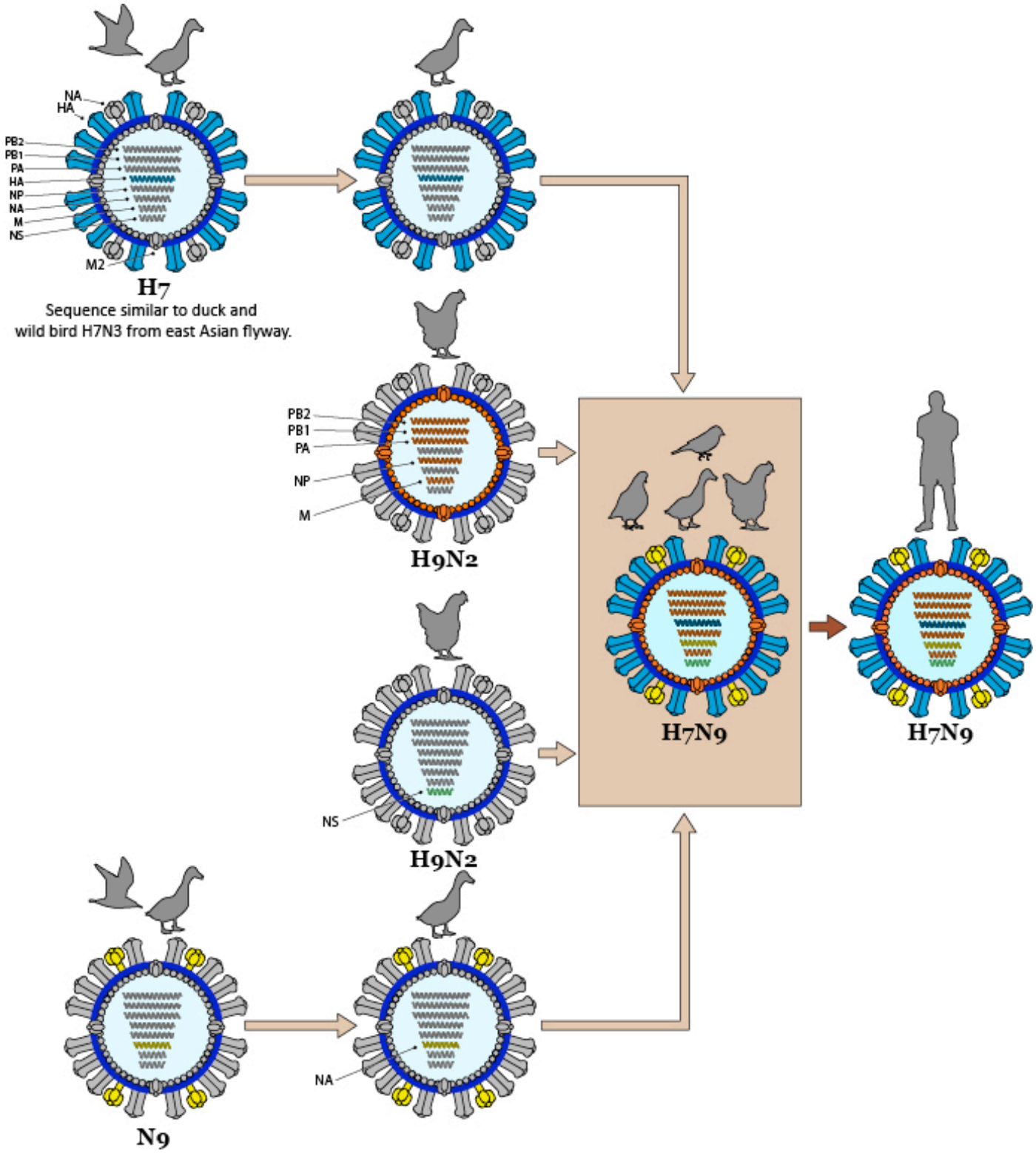


Fig. 4.2

(e) Within reference to **Fig. 4.2**, explain how the new combination of RNA segments in H7N9 could have arisen. [2]

- **Antigenic shift**;
- Where **two influenza strains** (one carrying the H7 and one carrying the N9) **infect the same intermediate host cell** at the **same time**;
- The cell is **replicating the viral genomes** of the **two strains** at the same time;
- **Random assembly of RNA segments** from the three different influenza viruses - **Ressortment of genes** occur resulting in the **new combination of RNA segments in influenza virus H7N9** when it exits by budding;

[Total: 11]

QUESTION 5

Carbon is essential to life. All of our molecular machines are built around a central scaffolding of organic carbon. Unfortunately, carbon in the earth and atmosphere is locked in highly oxidized forms, such as carbonate minerals and carbon dioxide gas. In order to be useful, this oxidized carbon must be "fixed" into more organic forms, rich in carbon-carbon bonds and decorated with hydrogen atoms. Powered by the energy of sunlight, plants perform this central task of carbon fixation.

Inside plant cells, the enzyme ribulose bisphosphate carboxylase (rubisco) forms the bridge between life and the lifeless, creating organic carbon from the inorganic carbon dioxide in the air. **Fig. 5.1** shows the structure of ribulose bisphosphate carboxylase (rubisco).

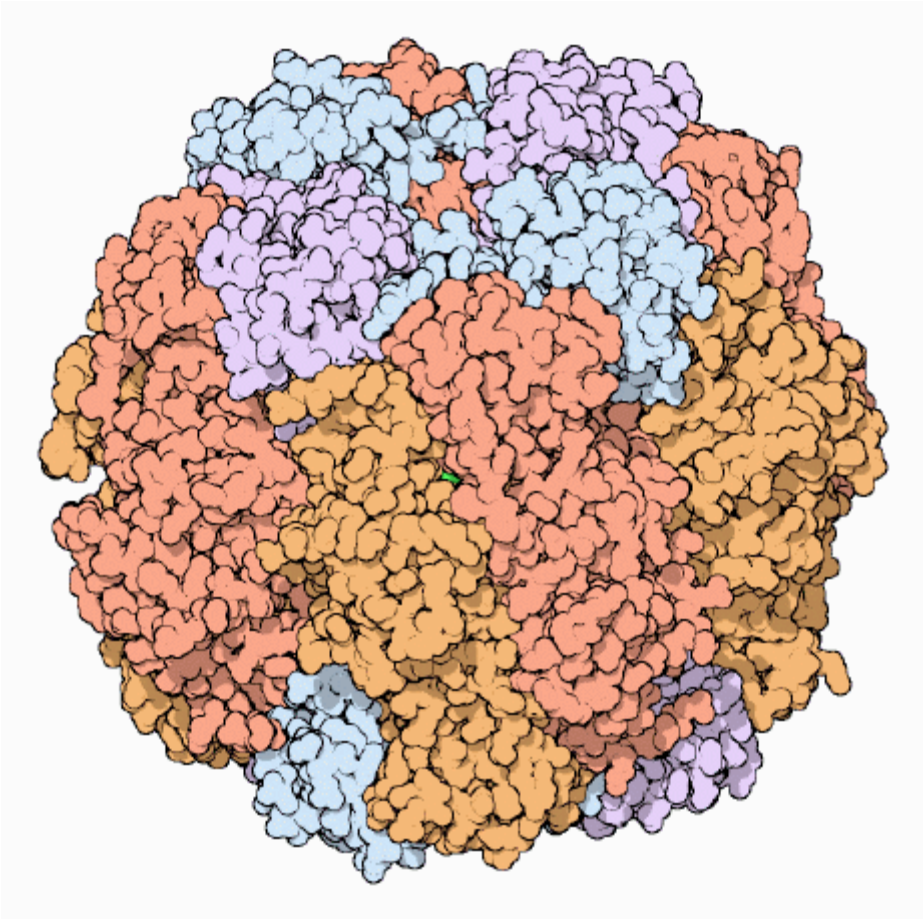


Fig. 5.1

(a) Rubisco is an enzyme made of sixteen polypeptide chains.

Describe how the structure of rubisco is maintained. [2]

- The primary sequence of the 16 polypeptide chains formed by having peptides held between the amino acids
- The folding of the individual polypeptides to form secondary structures (Alpha helices and Beta-pleated sheets) and maintained by H bonds between the imino and carbonyl group.
- These secondary structures will form tertiary structures with further compacting and folding and these are held by hydrogen bonds, hydrophobic interaction, ionic bonds and disulphide bonds.
- These tertiary structures will associate with each other to form rubisco with each tertiary structure held by the same interactions as well.

In spite of its central role, rubisco is remarkably inefficient. Typical enzymes can process a thousand molecules per second, but rubisco fixes only about three carbon dioxide molecules per second. **Fig. 5.2** shows the relationship between CO_2 assimilation rate and increasing light intensity in a plant, when carbon dioxide is present in excess.

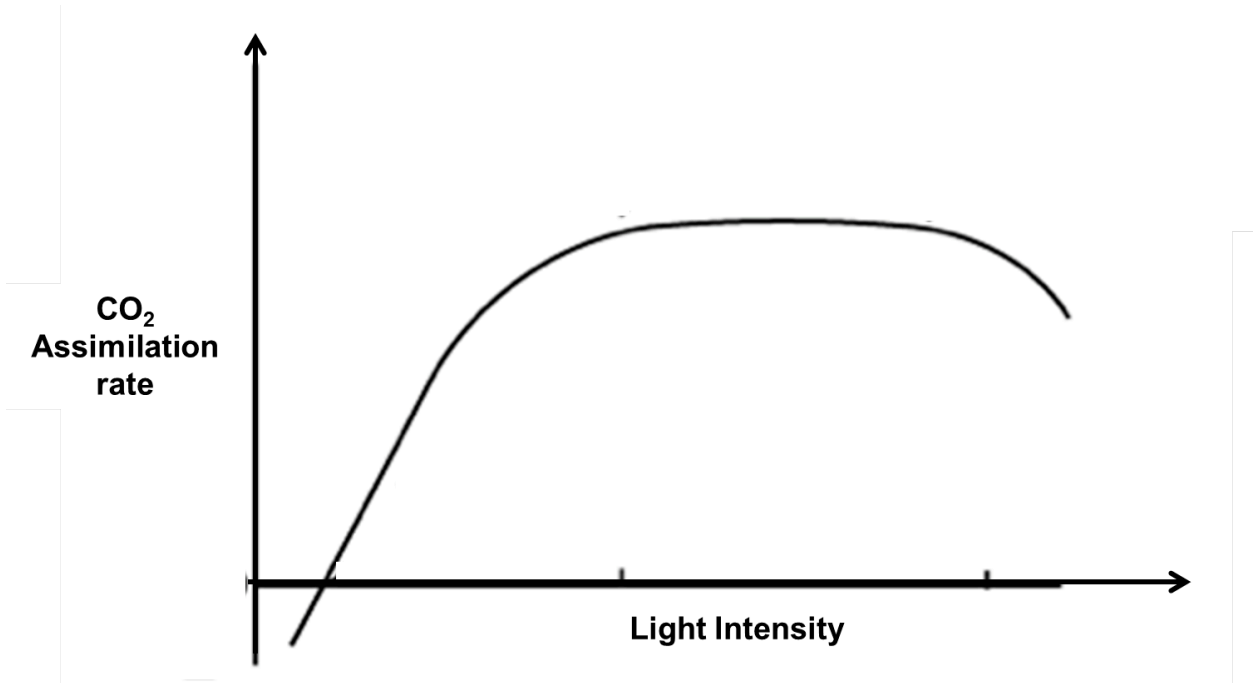


Fig. 5.2

(b) Explain how light energy assists in the process of carbon fixation in plants. [3]

- Light energy is harvested by the antenna complex and causes excitation of electrons in the photosystem I and II
- Put through the process of photophosphorylation
- The excitation of electron of the PS I and II is captured by the primary electron acceptors of each PS
- This electron from PSII is passed down an electron transport chain to generate ATP through chemiosmosis
- The ATP formed will be used during the Calvin cycle during regeneration of RuBP
- In a second ETC, NADP is reduced by NADP reductase to form reduced NAD -> needed for Calvin cycle as a reducing power.

(c) Explain why the CO_2 assimilation rate levels off at higher light intensity. [3]

- CO_2 assimilation rate levels off at high light intensity →maximum rate as all photosystems are saturated with light energy (limited amount of chlorophyll in the photosystems)
- Maximum rate of photosynthesis occurs at high LI hence amount of CO_2 assimilation now remained constant due to the constant (maximum) amount of NADPH produced;
- Light is no longer limiting factor, but other factor e.g. rate of carbon fixation is.

Rubisco also shows an embarrassing lack of specificity. Unfortunately, oxygen molecules and carbon dioxide molecules are similar in shape and chemical properties. In rubisco, an oxygen molecule can bind comfortably in the active site designed to bind carbon dioxide. Rubisco then attaches the oxygen to the sugar chain, forming a faulty oxygenated product.

(c) With reference to **Fig 5.2**, suggest why the CO_2 assimilation rate is declining as the light intensity increases higher. [2]

- As light intensity gets higher, the rate of production of oxygen from photolysis of water to replace the lost of electrons in PS II may increase.
- Oxygen produced within the plant cell will be greater than amount of carbon dioxide present and Oxygen will bind rubisco preferentially and thereby leading to a decline in CO_2 assimilation

[Total: 10]

QUESTION 6

Hair color is the pigmentation of hair follicles due to two types of melanin: eumelanin and pheomelanin. Pheomelanin colors hair reddish-orange. Eumelanin, which has two subtypes of black or brown, determines the darkness of the hair color. Generally, if more eumelanin is present, the color of the hair is darker. If less eumelanin is present, the hair is lighter. Particular hair colors are associated with ethnic groups.

The genetics of hair colors are not yet firmly established. According to one theory, at least two unlinked gene pairs that control human hair color.

In Americans, gene **B** and **R** are involved in controlling the shades of hair colour. Gene **B** controls the production of brown eumelanin. This gene has a dominant brown allele and a recessive blond allele. Gene **R** controls the production of pheomelanin. This gene has a non-red allele (which suppresses production of pheomelanin) which is dominant and the allele for red hair is recessive. Gene **B** is epistatic over Gene **R**.

(a) State and explain the genotypes for a blonde hair phenotype. [3]

- bbRr and bbRR [1]
- to have blonde hair the person must be homozygous recessive for gene B (bb) such that the will be no production of brown eumelanin.
- At the loci for Gene R should contain at least a copy of the dominant allele R to suppress production of pheomelanin to tint the hair red.

A couple, both with blonde hair, was filing for a divorce as the husband felt that her wife has cheated on him. His rationale of saying so was based on the recent birth of their boy who has red-coloured hair.

(b) With the use of a genetic diagram, explain whether or not it is possible for two parents with blonde hair to have an offspring with red-coloured hair. [4]

	Woman		**Man**	
Parental phenotype	Blonde Hair	x	Blonde Hair	1 m
Parental genotype	**bbRr**		**bbRr**	
Gametes	**bR** **br**		**bR** **br**	1 m

Offspring phenotype (fertilization lines or punnett square needs to be shown) [1] for correct genotype and phenotype

	bR	**br**
bR	**bbRR** Blonde hair	**bbRr** Blonde Hair
br	**bbRr** Blonde hair	**bbrr** Red hair

1 m

Offspring genotypic ratio	**bbRR**	:	**2 bbRr**	:	**bbrr**	:	1 m
Offspring phenotypic ratio	Blonde hair	:	Blonde hair	:	Red hair	:	

- Possible as the both parents might be heterozygous for the loci of Gene R.

(c) The husband felt bad for doubting his wife. He wants to find our whether there is a significant difference between what they observed in their baby boy and what is expected.

Explain whether it is possible to carry out a chi-square test in this case. [2]

- **Not possible [1]**
- No large sample size of offsprings to conduct a chi-squared test.

[Total: 9]

QUESTION 7

Fig. 7.1 shows Honeycreepers of the Family Drepanididae from the Hawaiian Islands. These Honeycreepers reside on the various Hawaiian Islands. Through a series of studies, these Honeycreepers are found to be evolved from an ancestral Honeycreeper which resided originally on, Kauai, one of the Hawaiian Islands.

Fig. 7.1

(a) Explain how these various species of Honeycreepers could have arisen. [4]

Mutation

- Spontaneous mutation or inherited variation in the genotype of these organisms provided genetic variation of each organism

Natural Selection

- Difference in selection pressure (e.g. food type) due to the different habitat that these forms are found.
- There would be one phenotype that would be at a selective advantage in the environment as it is better adapted to the living conditions. They will survive, reproduce and pass on their genes to offsprings.

Isolation

- As these different forms are found on the different islands/ habitats, they are geographically isolated. Interbreeding is prevented and hence, prevention of gene flow.

Genetic drift

- Also, the common ancestor might be genetically different in the natural population in the area of origin. When they migrate to a new island, there will be Founder effect. As a result of the loss of genetic variation, the new population may be distinctively different, both genetically and phenotypically, from the parent population from which it is derived.

Our understanding of the phylogeny of species has been greatly improved through the use of molecular techniques.

Many phylogenetic studies use mitochondrial DNA rather than nuclear DNA.

(b) Explain what is meant by the term phylogeny. [2]

- **Phylogeny** is the organisation of species according to particular characteristics
- which takes into consideration the evolutionary relationship between the species.

A new Honeycreeper (**Fig. 7.2**) has been found on one of the Hawaiian Islands. Based on its structural features, Scientists are speculating that this new found Honeycreeper is also a descendant of the ancestral Honeycreeper found on Kauai.

Fig. 7.2

(c) Suggest why it is not advisable to determine the phylogenetic relationship of this new found Honeycreeper to the rest of them from the morphological features. [2]

- Basing on morphological features is subjective and non-quantifiable.
- Similarities in morphological structures could be due to convergent evolution

(d) Outline how DNA can be studied to determine phylogeny of organisms. [5]

Studying DNA from different organisms is working on the principle of molecular homology

- DNA samples can be extracted from target organisms or from the bone marrow of fossils. With techniques of DNA sequencing technology, the DNA can be sequenced.
- For basis of valid comparison the sequence of similar genes (for example, cytochrome c oxidase gene) are compared amongst the organisms.
- A data base of DNA sequences of genes could also be used to retrieve information of particular organism of interest.
- Degree of similarity in sequences/ degree which sequences of gene are conserved between species are interpreted as DNA evidence for evolution.
- Generally, a higher degree which sequences are being conserved a higher degree of relatedness different species are to one another and vice versa.

[Total: 13]

SECTION B (20 MARKS)

Answer **one** question.
Write your answers on the separate writing paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose.
Your answers must be set out in sections (a), (b) etc., as indicated in the question.

QUESTION 8

(a) Compare anaerobic respiration and aerobic respiration. [6]

Max 3 for differences	**Anaerobic Respiration**	**Aerobic Respiration**
Definition/ Components	Glycolysis + Fermentation	Glycolysis + Link Reaction + Krebs Cycle + Oxidative phosphorylation
Oxygen requirement	Absence of Oxygen	Presence of Oxygen
ATP production	2 ATP	38 ATP
Electron carriers involved	NAD only	NAD and FAD
Location	Cytoplasm only	Cytoplasm and Mitochondria
Similarities [3 marks]		
• Starting materials is glucose • Both has glycolysis as a step in the process of oxidation of glucose • Both involves the use of electron carriers (NAD or FAD) • Carbon dioxide gas is produced in both processes		

(b) Describe the main features of oxidative phosphorylation. [8]

1) Occurs in the inner mitochondrial membrane
2) Synthesis of ATP occurs as electrons are transferred from reduced NAD and reduced FAD generated from the Krebs cycle, link reaction or glycolysis
3) Down the Electron Transport Chain (ETC) found on the inner mitochondrial membrane
4) The energy released from the electron transport chain is used to actively pump protons (H^+) from the matrix to the intermembrane space of the mitochondria.
5) This thus creates a proton gradient across the inner membrane. The proton gradient serves as a source of potential energy for the synthesis of ATP.
6) The stalked particles on the inner mitochondrial membrane consist of hydrophilic protein channels called F_0 proteins and the enzyme ATP synthase.
7) As H^+ diffuses through the F_0 proteins into the matrix of the mitochondria, the energy released by the protons is used by ATP synthase to attach an inorganic phosphate to ADP, thus forming ATP. The electrochemical gradient is said to be coupled to ATP synthesis.
8) Oxidation of one molecule of NADH yields 3 ATP while oxidation of one molecule of $FADH_2$ yields 2 ATP by oxidative phosphorylation.

9) The electrons are passed down the electron transport chain. O_2 the final electron acceptor is reduced to H_2O.

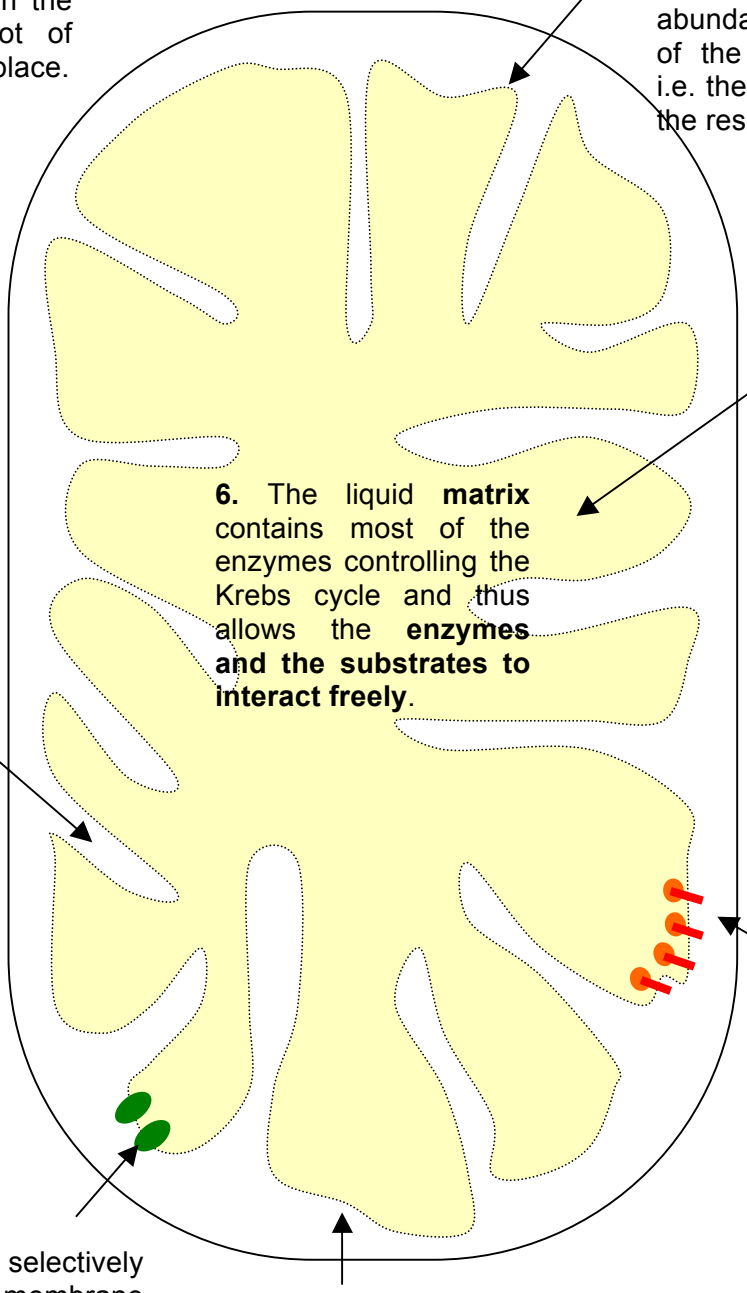
10) OP allows regeneration of NAD and FAD for the use the other components of respiration.

(c) Discuss how the mitochondrion is adapted for carrying out respiration. [6]

Fig. 9 Adaptation of mitochondrion to aerobic respiration

8. Mitochondria can **move** to areas in the cell where a lot of activity is taking place.

1. The **cristae** of the inner membrane effectively **increase** its **surface area** to provide abundant space for the layout of the multi-enzyme systems. i.e. the sequence of carriers for the respiratory chain.



2. The immediate **bathing of the cristae by the matrix** and the great surface area provided by the cristae allow the carriers **greater access** to the substrates found in the matrix.

6. The liquid **matrix** contains most of the enzymes controlling the Krebs cycle and thus allows the **enzymes and the substrates to interact freely**.

7. Mitochondria in **active** cells are **larger** and contain **more cristae**.

3. Elementary particles associated with ATP synthesis are found on the **matrix side** of the inner membrane

5. The selectively permeable membrane contains translocase enzymes for the **active transport of ATP and ADP through it.**

4. The **inner membrane is impermeable to protons** moving in from the matrix. A proton pump in the inner membrane pumps protons from matrix into intermembrane space.

The **outer membrane is freely permeable to protons** and hence allows protons to move out of the mitochondria. Protons accumulate outside the mitochondria and create a pH gradient. The pH gradient allows protons to diffuse back into the mitochondrion in which ATP is synthesized during the process.

QUESTION 9

(a) Compare the mode of infection of Influenza Virus and Human Immunodeficiency Virus. [6]

3 Differences	Influenza Virus	HIV
Adsorption	With the use of HA	With the use of GP 120
Penetration	By Receptor mediated endocytosis	Fusion of Viral envelope with host cell membrane
Reverse transcription	No reverse transcription of Viral RNA genome	Reverse transcription of viral RNA genome to for Viral DNA
Integration	No integration of Viral genome into host cell genome	Integration of Viral DNA to host cell genome by integrase
Similarities (3 Similarities)		
• Nucleocapsid are both removed enzymatically • Both viruses involve embedding viral glycoproteins on the host cell membrane for the formation of viral envelope • Host cell ribosomes are used for the translation of viral polypeptide for the formation of viral particles.		

(b) Describe the mechanisms of gene transfer in prokaryotes. [8]

- There are three mechanisms of gene transfer in prokaryotes, Conjugation, Transformation and Transduction.

Conjugation [max 2 marks]

- Ref. to mating signal
- F+ cell (cell containing the F Plasmid) trigger the formation of sex pillus – act as a mating bridge between donor and recipient.
- Plasmid from the donor cell replicates via the Rolling Circle Replication – a single strand of DNA enters the recipient cell through the sex pillus
- Plasmid in recipient reforms its double stranded structure -> recipient cell acquire genes from donor cell.

Transformation [max 3 marks]

- Bacteria cells must be competent (possesses competence-specific proteins)
- DNA from the environment binds to these competence-specific proteins
- Autolysin are present to degrade the cell wall to allow DNA to penetrate through cell wall of bacterium
- Nucleases present then cut the bound DNA into fragments -> single strand of DNA
- Single- stranded DNA will remain bound by competence-specific protein and be brought near bacterial chromosome
- RecA protein will take over to promote genetic exchange between the DNA fragment with the recipient bacterial chromosome.

Transduction [max 3 marks]

- Transduction is a genetics exchange method between bacterial cells facilitated by bacteriophages
- Two mode of operation: specialised transduction and generalised transduction
- By T4 phage and Lambda phage respectively
- During the reproductive cycle of these phages, there is incorporation of bacterial genes into the phage capsid during formation of mature phages
- These bacteriophages with the incorporated bacterial DNA will subsequently infect a new bacterial host cell
- Genetic recombination will occur between the bacterial genes from the previous host cell and the genes in the new bacterial cell.

(c) Discuss how the prokaryote, E. coli, is able to respond to varying concentrations of lactose. [6]

- The prokaryotes possess lac operon which contains the structural genes lac Z, lac Y and lac A genes
- Which codes for beta-galactosidase, lactose permease and lactose transacetylase respectively
- These structural genes are organised under the control of a single promoter and operator
- Expression of these genes are regulated depending on the level of lactose in the environment.
- When [lactose] low, some lactose is transported by the low levels of permease into bacterial cells
- Lactose converts to allolactose by beta-galactosidase
- Allolactose binds to lac repressor to induce a change in conformation in repressor
- Repressor unable to bind to operator
- RNA polymerase can bind promoter to initiate transcription
- When [lactose] high, more repressors are repressed by allolactose
- These are occurring in absence of glucose, when glucose present -> glucose prevent synthesis of cAMP
- cAMP needed for activation of CAP
- CAP needed to bind to CAP binding site for the recruitment of RNA polymerase to initiate transcription.

End of Paper